Exercises Preparation



Install and Setup Hive

Create VM Instance

1. Delete previously created VM instance:

```
gcloud compute instances delete big-data
```

2. Create new instance:

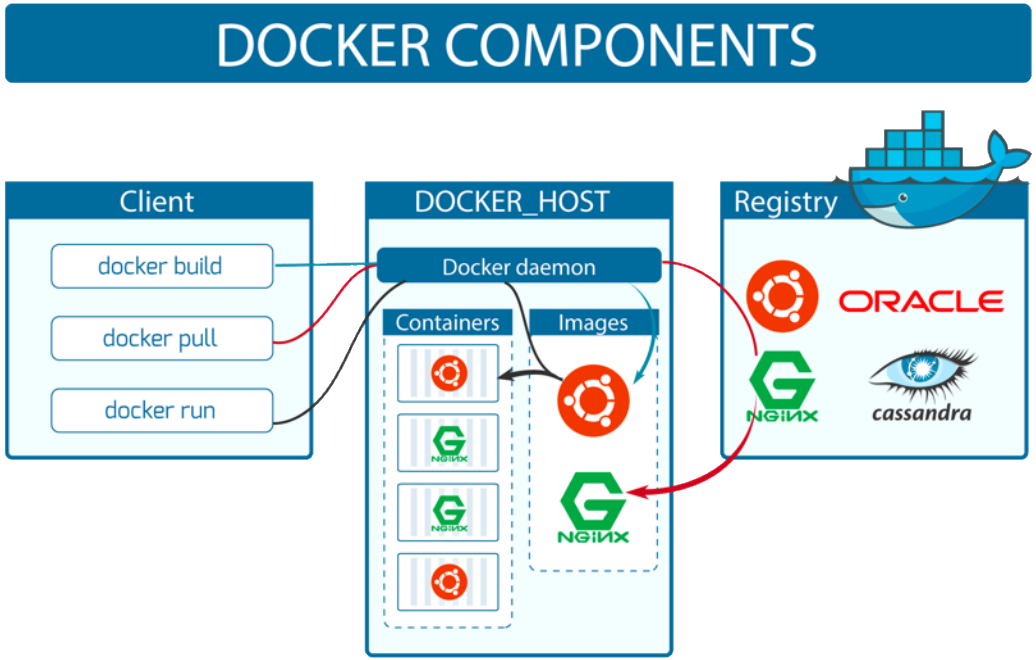
```
gcloud compute --project=[your-project-id] instances create big-data \
          --zone=europe-west3-a \
          --machine-type=n2-highmem-4 \
          --subnet=default --network-tier=PREMIUM \
          --maintenance-policy=MIGRATE \
          --image=ubuntu-2004-focal-v20210129 \
          --image-project=ubuntu-os-cloud \
          --boot-disk-size=100GB \
          --boot-disk-type=pd-standard \
          --boot-disk-device-name=big-data
```

```
ssh hans.wurst@XXX.XXX.XXX.XXX
```

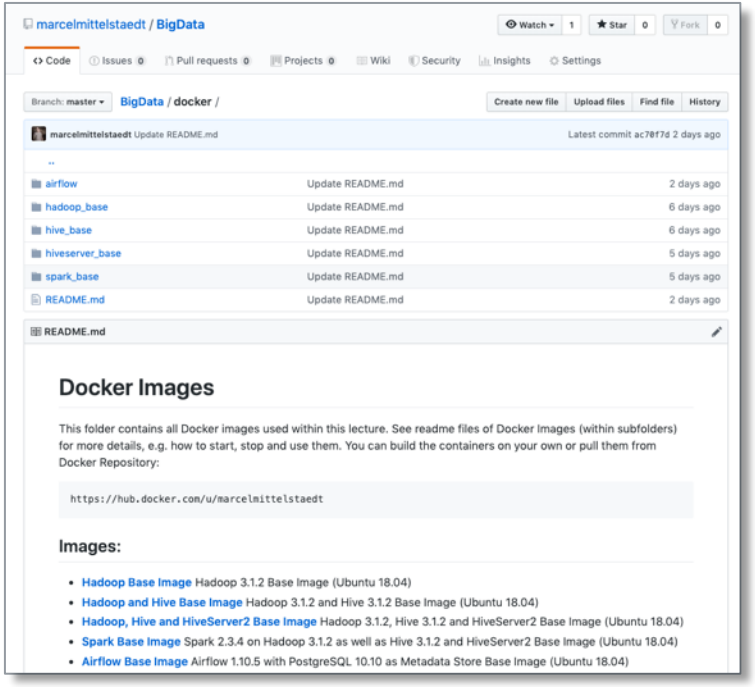



Docker

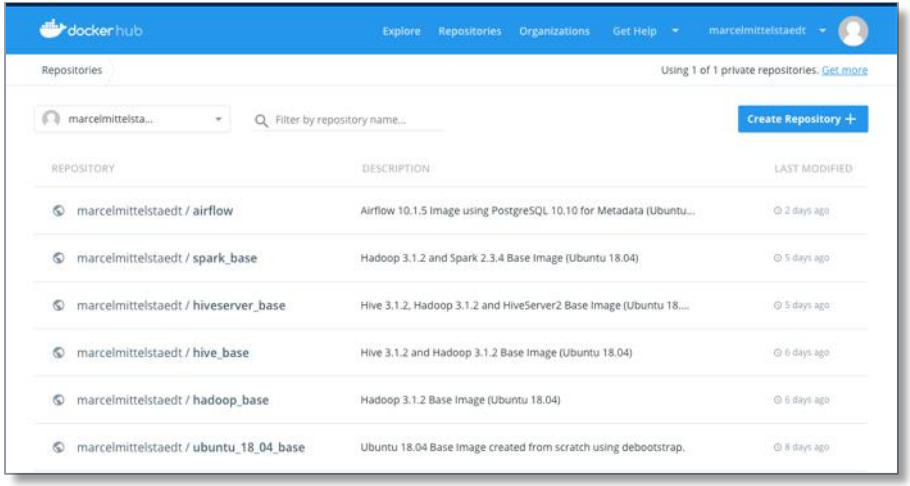
To speed things up and not waste time on installation and configuration of Hive and other tools, we will make use of docker container I‘ve already prepared.



Docker Images/Dockerfiles



https://github.com/marcelmittelstaedt/BigData/tree/master/docker



https://hub.docker.com/u/marcelmittelstaedt

Setup Docker Container

3. Install and setup docker

```
sudo apt-get update
sudo apt-get install docker.io
sudo usermod -aG docker $USER
# exit and login again
```

4. Pull Hadoop with Hive Image

```
docker pull marcelmittelstaedt/hive_base:latest
```

5. Start Container from pulled image:

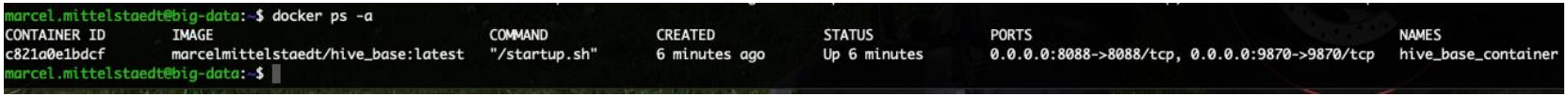
```
docker run -dit --name hive_base_container -p 8088:8088 -p 9870:9870 -p 9864:
9864 marcelmittelstaedt/hive_base:latest
```




Setup Docker Container

6. Show Running Container:

```
docker ps -a
```



7. Show Logs of container (wait till finished):

```
docker logs hive_base_container

[…]
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [c821a0e1bdcf]
Stopping nodemanagers
Stopping resourcemanager
Container Startup finished.
```

Setup Docker Container

8. Get a shell inside the container:

```
hans.wurst@big-data:~$ docker exec -it hive_base_container bash
root@c821a0e1bdcf:/#
```

9. Switch to hadoop user:

```
root@c821a0e1bdcf:/# sudo su hadoop
hadoop@c821a0e1bdcf:/$ cd
hadoop@c821a0e1bdcf:~$
```

10. Start DFS and YARN:

```
start-all.sh
```




Test Hive

11. Test if Hive Installation and Configuration is successful. Start Hive:

```
hive
```



```
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hadoop/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/
impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hadoop/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7
.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Hive Session ID = c120d0b1-9025-43db-96e4-48ccfb875f1a

Logging initialized using configuration in jar:file:/home/hadoop/hive/lib/hive-common-3.1.0.jar
!/hive-log4j2.properties Async: true
Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider us
ing a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
Hive Session ID = fdd6f06a-d4e4-48e6-8971-997e8a0a8e2c
hive>
```


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Install and Setup Hive

12. Execute First SQL Query:

```
hive> show databases;
OK
default
Time taken: 0.083 seconds, Fetched: 1 row(s)
hive>
```




Break

TIME FOR A BREAK



Hive: Create and Work with External Tables

Using public dataset of IMDb.com



Get IMDb Data And Move It To HDFS

1. Get **IMDb Data** (*https://www.imdb.com/interfaces/*):

```
wget https://datasets.imdbws.com/title.basics.tsv.gz
wget https://datasets.imdbws.com/title.ratings.tsv.gz
```

2. Uncompress IMDb Data:

```
gunzip title.basics.tsv.gz
gunzip title.ratings.tsv.gz
```

3. Create HDFS Directories for IMDb Data:

```
hadoop fs -mkdir /user/hadoop/imdb
hadoop fs -mkdir /user/hadoop/imdb/title_basics
hadoop fs -mkdir /user/hadoop/imdb/title_ratings
```



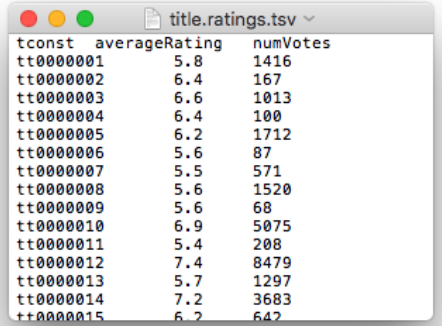

Create External Tables In Hive

4. Transfer IMDb data files to HDFS:

```
hadoop fs -put title.basics.tsv /user/hadoop/imdb/title_basics/title.basics.tsv
hadoop fs -put title.ratings.tsv /user/hadoop/imdb/title_ratings/title.ratings.tsv
```

5. Create External Table **`title_ratings`** (file *title.ratings.tsv*) in Hive:

```
hive > CREATE EXTERNAL TABLE IF NOT EXISTS title_ratings(
          tconst STRING,
          average_rating DECIMAL(2,1),
          num_votes BIGINT
) COMMENT 'IMDb Ratings'
ROW FORMAT DELIMITED FIELDS TERMINATED BY '\t' STORED AS
TEXTFILE LOCATION '/user/hadoop/imdb/title_ratings'
TBLPROPERTIES ('skip.header.line.count'='1');
```



Create External Tables In Hive

6. Create External Table `title_basics` for file *title.basics.tsv* in Hive:

title.basics.tsv

tconst	titleType	primaryTitle	originalTitle	isAdult	startYear	endYear	runtimeMinutes	genres
tt0000001	short	Carmencita	Carmencita	0	1894	\N	1	Documentary,Short
tt0000002	short	Le clown et ses chiens	Le clown et ses chiens	0	1892	\N	5	Animation,Short
tt0000003	short	Pauvre Pierrot	Pauvre Pierrot	0	1892	\N	4	Animation,Comedy,Romance
tt0000004	short	Un bon bock	Un bon bock	0	1892	\N	\N	Animation,Short
tt0000005	short	Blacksmith Scene	Blacksmith Scene	0	1893	\N	1	Comedy,Short
tt0000006	short	Chinese Opium Den	Chinese Opium Den	0	1894	\N	1	Short
tt0000007	short	Corbett and Courtney Before the Kinetograph	Corbett and Courtney Before the Kinetograph	0	1894	\N	1	Short,Sport
tt0000008	short	Edison Kinetoscopic Record of a Sneeze	Edison Kinetoscopic Record of a Sneeze	0	1894	\N	1	Documentary,Short
tt0000009	movie	Miss Jerry	Miss Jerry	0	1894	\N	45	Romance
tt0000010	short	Employees Leaving the Lumière Factory	La sortie de l'usine Lumière à Lyon	0	1895	\N	1	Documentary,Short
tt0000011	short	Akrobatisches Potpourri	Akrobatisches Potpourri	0	1895	\N	1	Documentary,Short
tt0000012	short	The Arrival of a Train	L'arrivée d'un train à La Ciotat	0	1896	\N	1	Documentary,Short
tt0000013	short	The Photographical Congress Arrives in Lyon	Neuville-sur-Saône: Débarquement du congrès des photographes à Lyon	0	1895	\N	1	Documentary,Short
tt0000014	short	Tables Turned on the Gardener	L'arroseur arrosé	0	1895	\N	1	Comedy,Short
tt0000015	short	Autour d'une cabine	Autour d'une cabine	0	1894	\N	2	Animation,Short
tt0000016	short	Barque sortant du port	Barque sortant du port	0	1895	\N	1	Documentary,Short
tt0000017	short	Italienischer Bauerntanz	Italienischer Bauerntanz	0	1895	\N	1	Documentary,Short
tt0000018	short	Das boxende Känguruh	Das boxende Känguruh	0	1895	\N	1	Short
tt0000019	short	The Clown Barber	The Clown Barber	0	1898	\N	\N	Comedy,Short
tt0000020	short	The Derby 1895	The Derby 1895	0	1895	\N	1	Documentary,Short,Sport
tt0000022	short	Blacksmith Scene	Les forgerons	0	1895	\N	1	Documentary,Short
tt0000023	short	The Sea Baignade en mer	0	1895	\N	1	Documentary,Short	
tt0000024	short	Opening of the Kiel Canal	Opening of the Kiel Canal	0	1895	\N	\N	News,Short
tt0000025	short	The Oxford and Cambridge University Boat Race	The Oxford and Cambridge University Boat Race	0	1895	\N	\N	News,Short,Sport
tt0000026	short	The Messers. Lumière at Cards	Partie d'écarté	0	1896	\N	1	Documentary,Short
tt0000027	short	Cordeliers' Square in Lyon	Place des Cordeliers à Lyon	0	1895	\N	1	Documentary,Short
tt0000028	short	Fishing for Goldfish	La pêche aux poissons rouges	0	1895	\N	1	Documentary,Short
tt0000029	short	Baby's Dinner	Repas de bébé	0	1895	\N	1	Documentary,Short
tt0000030	short	Rough Sea at Dover	Rough Sea at Dover	0	1895	\N	1	Documentary,Short

Create External Tables In Hive

6. Create External Table `title_basics` for file *title.basics.tsv* in Hive:

```
hive > CREATE EXTERNAL TABLE IF NOT EXISTS title_basics (
    tconst STRING,
    title_type STRING,
    primary_title STRING,
    original_title STRING,
    is_adult DECIMAL(1,0),
    start_year DECIMAL(4,0),
    end_year STRING,
    runtime_minutes INT,
    genres STRING
) COMMENT 'IMDb Movies' ROW FORMAT DELIMITED FIELDS TERMINATED BY
'\t' STORED AS TEXTFILE LOCATION '/user/hadoop/imdb/title_basics'
TBLPROPERTIES ('skip.header.line.count'='1');
```

Create External Tables In Hive

7. Query Table `title_basics` in Hive using SQL (HiveQL):

```
hive> select * from title_basics limit 3;
OK
tt0000001 short Carmencita Carmencita 0 1894 NULL 1 Documentary,Short
tt0000002 short Le clown et ses chiens Le clown et ses chiens 0 1892 NULL 5 Animation,Short
tt0000003 short Pauvre Pierrot Pauvre Pierrot 0 1892 NULL 4 Animation,Comedy,Romance
Time taken: 0.139 seconds, Fetched: 5 row(s)
hive>
```

8. Query Table `title_ratings` in Hive using SQL (HiveQL):

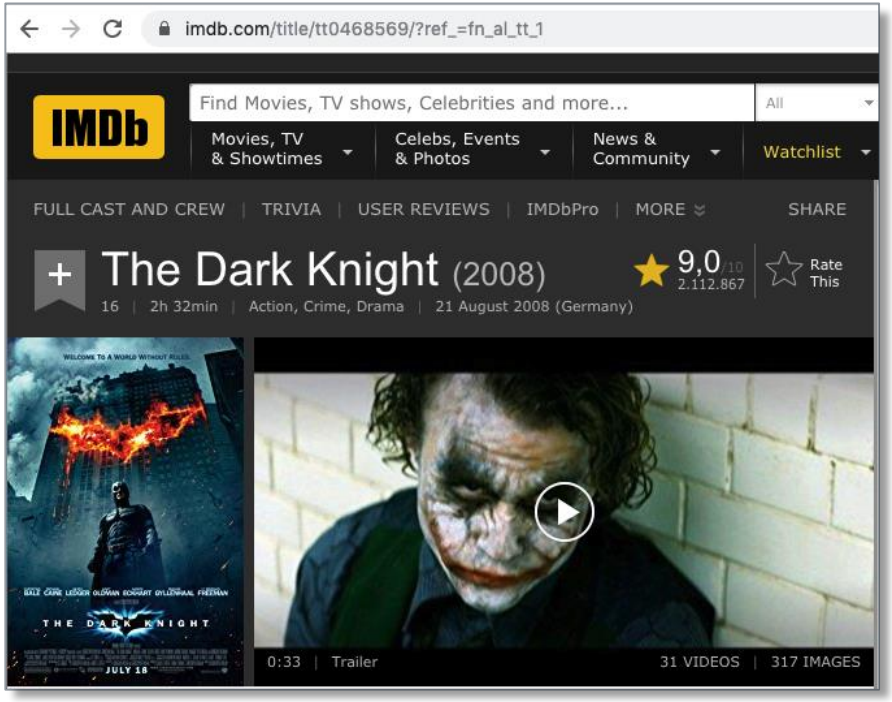
```
hive> select * from title_ratings limit 3;
OK
tt0000001 5.6 1540
tt0000002 6.1 186
tt0000003 6.5 1199
Time taken: 0.119 seconds, Fetched: 3 row(s)
hive>
```




Create External Tables In Hive

9. Run a complex query which starts a MapReduce Job on Yarn, e.g. get Rating of movie „*The Dark Knight*“:

```
SELECT
   *
FROM
   title_basics b
   JOIN title_ratings r ON (b.tconst=r.tconst)
WHERE
   original_title = 'The Dark Knight'
   AND title_type='movie';
```



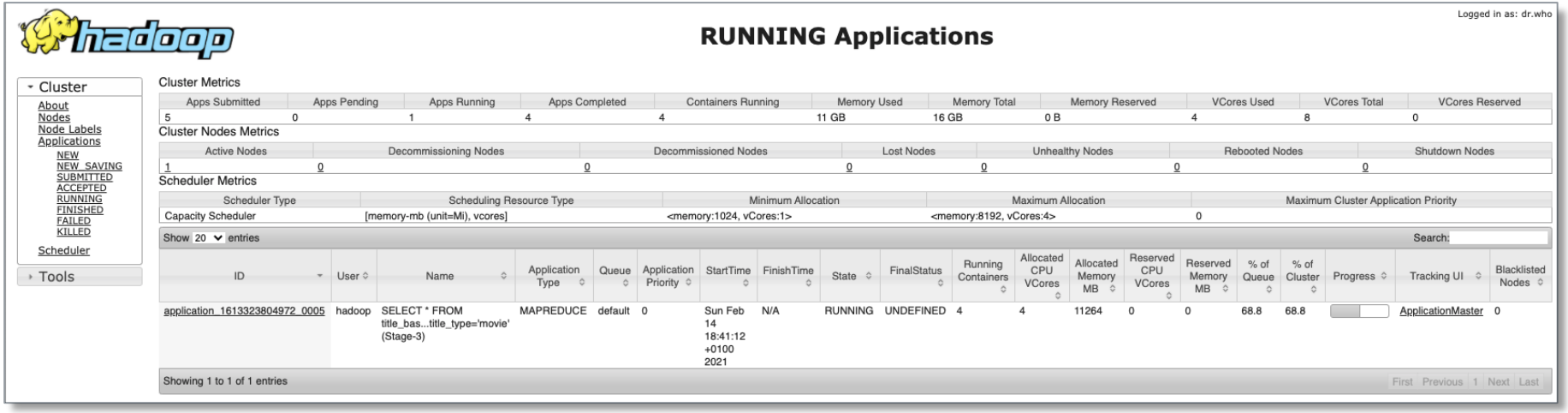
Create External Tables In Hive

9. Execute Query

```
hive> SELECT * FROM title_basics b JOIN title_ratings r ON (b.tconst=r.tconst) WHERE original_title =
'The Dark Knight' and title_type='movie';
[…]
Starting Job = job_1613323804972_0003, Tracking URL = http://154172c92bc7:8088/proxy/application_1613323804972_0003/
Kill Command = /home/hadoop/hadoop/bin/mapred job  -kill job_1613323804972_0003
Hadoop job information for Stage-3: number of mappers: 3; number of reducers: 0
2021-02-14 17:37:59,326 Stage-3 map = 0%,  reduce = 0%
2021-02-14 17:38:20,617 Stage-3 map = 50%,  reduce = 0%, Cumulative CPU 35.59 sec
2021-02-14 17:38:21,656 Stage-3 map = 67%,  reduce = 0%, Cumulative CPU 52.65 sec
2021-02-14 17:38:22,718 Stage-3 map = 100%,  reduce = 0%, Cumulative CPU 56.84 sec
MapReduce Total cumulative CPU time: 56 seconds 840 msec
Ended Job = job_1613323804972_0003
MapReduce Jobs Launched:
Stage-Stage-3: Map: 3   Cumulative CPU: 56.84 sec   HDFS Read: 650217476 HDFS Write: 376 SUCCESS
Total MapReduce CPU Time Spent: 56 seconds 840 msec
OK
tt0468569 movie The Dark Knight The Dark Knight 0 2008 NULL 152 Action,Crime,Drama tt0468569 9.0 2111245
Time taken: 53.094 seconds, Fetched: 1 row(s)
hive>
```

Create External Tables In Hive

9. Take a look at YARN (http://XXX.XXX.XXX.XXX:8088/cluster/):



Break

TIME FOR A BREAK





Exercises

Hive: Create and Work with External Tables

HDFS and HiveQL Exercises - **IMDB**

1. Execute Tasks of previous HandsOn Slides
2. Download https://datasets.imdbws.com/name.basics.tsv.gz
3. Create HDFS Directory **/user/hadoop/imdb/name_basics**/ for file name.basics.tsv
4. Create External Hive Table `name_basics` for name.basics.tsv
5. Use HiveQL to answer following questions:
   a) How many **movies** and how many **TV series** are within the IMDB dataset?
   b) Who is the **youngest** actor/writer/… within the dataset?
   c) Create a list (`tconst, original_title, start_year, average_rating, num_votes`) of movies which are:
   - equal or newer than year 2010
   - have an average rating equal or better than 8,1
   - have been voted more than 100.000 times

   d) How many movies are in list of c)?



HDFS and HiveQL Exercises - **IMDB**

5. Use HiveQL to answer following questions:

e) We want to know which years have been great for cinema.
Create a list with one row per year and a related count of movies which:
- have an average rating better than 8
- have been voted more than 100.000 times
ordered descending by count of movies.



Stop Your VM Instances

DON'T FORGET TO STOP YOUR VM INSTANCE!



```
gcloud compute instances stop big-data
```

